

Question 1 (20 marks)

qn	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
ans	B	A	D	D	D	A	D	A	B	A	A	D	B	B	C	D	A	D	D	A

- A servo motor is a typical example of
 - Electronics system
 - Mechanical system
 - Computer system
 - Mechatronics system
- The main function of Actuator is
 - To produce motion
 - Defect input
 - Detect output
 - Detect the state of the system
- Example of 'Stand-alone system' is
 - Machining centres
 - Washing machine
 - Robots for parts handling
 - Automated inspection stations
- Which among the following carry out the overall control of a system?
 - Graphical display
 - Sensors
 - Actuators
 - Digital controls
- Where is the feedback generated by sensors in a mechatronics system given?
 - Input sensors
 - Comparators
 - Mechanical actuators
 - Output sensors
- A Mechatronics system contains feedback.
 - True
 - False
- Which is the first aspect which needs to be considered in the Mechatronics design process?
 - Hardware integration and simulate
 - Conceptual design
 - Mathematical modeling
 - Modeling and simulation
- What converts physical input into output among the basic parts of a measuring system?
 - Transducer or sensor
 - Signal conditioning
 - Intelligence
 - Display
- Accuracy is the difference between a true value and the measured value.
 - True
 - False
- Which elements among the following is used to modify the data before display?
 - Data presentation element
 - Data transmission element
 - Data processing element
 - Variable manipulation element

11. What are the applications of PLC in mechatronics?

- Timing, counting, logic, arithmetic and sequencing
- Managing, commanding and directing
- Storing data
- Processing

12. What are transducers?

- They convert power from one form to another
- They convert work from one form to another
- They convert work to power
- They convert energy from one form to another

13. What is the full form of LVDT with respect to displacement transducer?

- Linear variable differential temperature
- Linear variable differential transformer
- Liquid visible differential transformer
- Liquified visible differential transformer

14. What is the open loop gain of an ideal operational amplifier?

- 0
- 1
- Infinite
- 1

15. What is the phase difference between input signal and output signal when input is provided to the inverting terminal of the operational amplifier?

- 0 degree
- 90 degrees
- 180 degrees
- 45 degree

16. What does DAQ stand for?

- Data Acquisition
- Data Acquisition quantity
- Data Acquisition quality
- Digital Acquisition

17. What is the magnitude of gain of an Op-Amp if its negative feedback resistance is 20 ohm and its input resistance is 10 ohm (consider the input is in inverting terminal)?

- 2
- 3
- 4
- 1/2

18. An example of discrete (digital) control is

- Varying the volume of a music system
- Turning a lamp ON or OFF
- Varying the brightness of a lamp
- Controlling the speed of a fan

19. Which of the following cannot be an input that is given to the PLC?

- a) Manual switches
- b) Relays
- c) Sensors
- d) None of the above

20. In a current sinking DC input module _____

- a) The current flows out of the input field device
- b) Requires that a AC sources be used with mechanical switches
- c) The current flows out of the input module
- d) Currents can flow in either direction at the input module

Question 2 (10 marks)

Operate the three phase AC motor forward and reverse as industrial process with all elements of mechatronics system. Draw the block diagram for this system.

